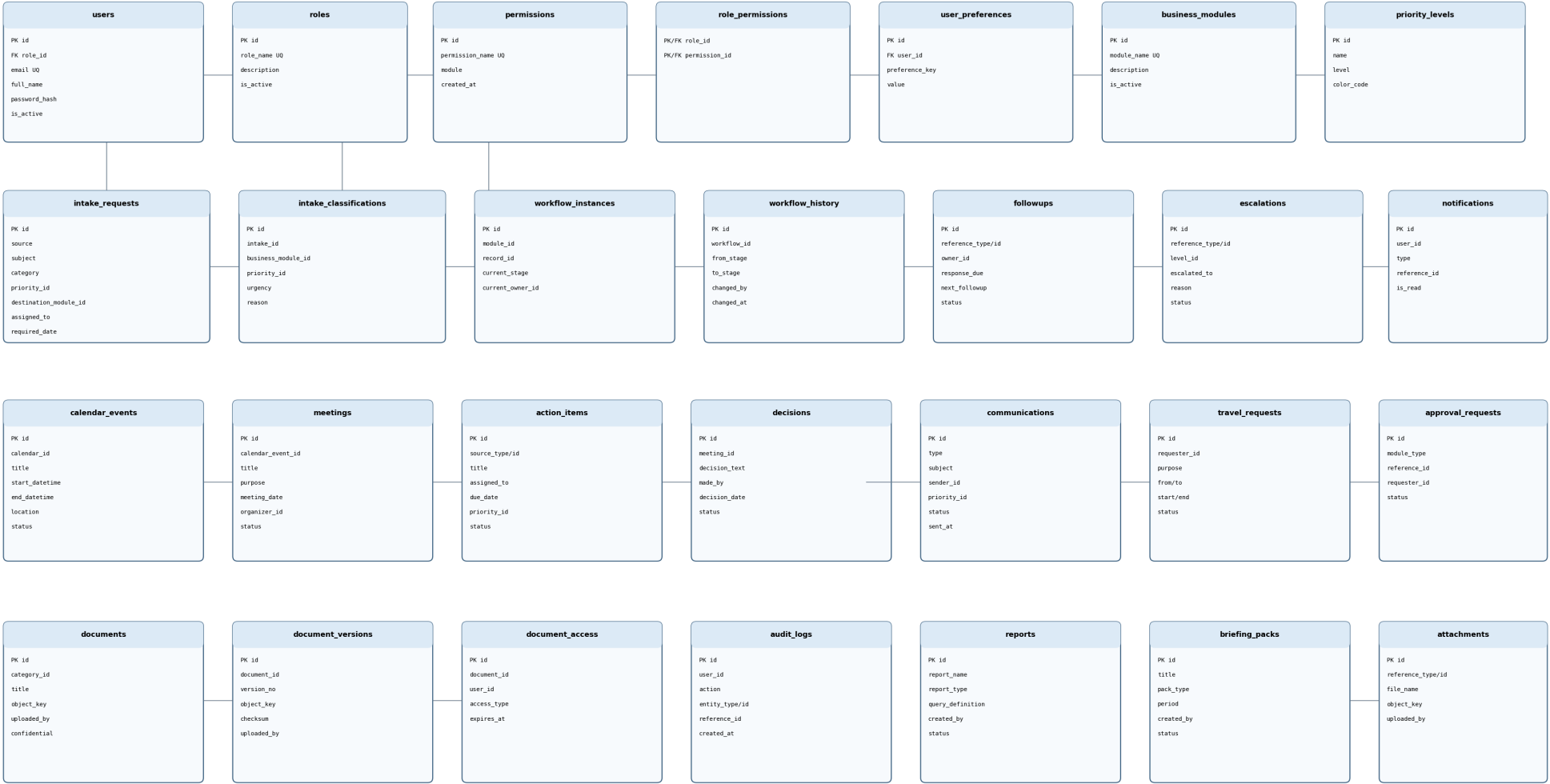


1. Source Architecture → Implementation Architecture

The uploaded specification defines **11 official business modules**. Intake & Classification, Common Workflow, Follow-up & Escalation, AI Assistance, Automation, Notifications, Audit/History, File Storage, Search/Relationships and Administration are shared supporting capabilities. The source maps Angular to UI/features, .NET to services/APIs/workflow/jobs/AI/audit, MySQL to structured records and DigitalOcean Spaces to actual files. ■filecite■turn0file0■L294-L313■

EA FMS — Proposed Logical ERD

Based on the uploaded 11-module EA FMS specification; table names/columns/relationships are an implementation proposal.



Shared engines: Intake → Classification → Module Record → Workflow → Follow-up/Escalation → Notifications → Audit. Files: DigitalOcean Spaces + MySQL metadata.

PK = Primary Key | FK = Foreign Key | UQ = Unique | For polymorphic reference_type/reference_id, enforce application-level referential integrity.

Angular	11 feature areas + shared UI, forms, dashboards, guards, API client and upload/download UI
.NET	Controllers, application/domain services, validation, workflow, escalation jobs, notifications, AI orchestration and audit
MySQL	Business records, relationships, workflow history, revisions, permissions, metadata and audit
DigitalOcean Spaces	Actual images/documents/files; MySQL stores metadata/object_key/access information
Background jobs	Due-date monitoring, reminders, escalation calculation, recurring work and report preparation

2. Recommended Delivery Roadmap

Phase	Build	Exit criteria
0 Foundation	Solution structure, environments, CI/CD, migrations, logging, Swagger	Angular + .NET + MySQL + Spaces running; dev/stage deployments
1 Security + Common	Users, roles, permissions, intake, priorities, attachments, audit	JWT/RBAC + intake CRUD/classification + common audit
2 Core I	Calendar, Communications, Documents	CRUD + search/filter + file metadata/version/access + calendar conflicts
3 Core II	Meetings, Actions, Decisions	Meeting→action/decision links; reusable workflow and history
4 Controlled Work	Approvals, Travel/Hospitality	Approval state machine + revisions + travel request/booking/itinerary/expense
5 Control Layer	Follow-up, reminders, L1–L5 escalation, notifications	Scheduled jobs reliably create reminders/escalations and resolution history
6 Management	Daily Workspace, Reports, Briefing Packs, dashboards	Daily/weekly/KPI/MIS/leadership outputs
7 AI Layer	AI assistance across intake, meetings, communication, decisions, follow-up, review, recommendations	Structured suggestions + human confirmation + audit
8 Hardening	Unit/integration/e2e/load/security/backup-restore/UAT	Production readiness checklist signed off

3. Shared Tables — Build These Before Module Tables

Table	Proposed logical columns / purpose
users	PK id; FK role_id; full_name; email; password_hash; phone; is_active; timestamps
roles	PK id; role_name; description; is_active
permissions	PK id; permission_name; module
role_permissions	PK/FK role_id + PK/FK permission_id
user_preferences	PK id; FK user_id; preference_key/value
business_modules	PK id; module_name; description; is_active
statuses	PK id; FK module_id; status_name; sequence; is_active
priority_levels	PK id; name; level; color_code; is_active
intake_requests	PK id; source; subject; description; category; priority_id; urgency; destination_module_id; assigned_to; required_date
intake_classifications	PK id; intake_id; business_module_id; priority_id; urgency; classification_reason
workflow_instances	PK id; module_id; record_id; current_stage; current_owner_id; started_at; completed_at
workflow_history	PK id; workflow_id; from_stage; to_stage; changed_by; remarks; changed_at
followups	PK id; reference_type/id; owner_id; response_due_date; last/next follow-up; status; notes
escalations	PK id; reference_type/id; escalation_level_id; escalated_to; reason; escalated_at; resolved_at; status
escalation_levels	PK id; level_name; level_no; sla_hours; is_active
notifications	PK id; user_id; title; message; type; reference_type/id; is_read; created_at
audit_logs	PK id; user_id; action; entity_type/id; reference_id; ip_address; created_at
attachments	PK id; reference_type/id; file_name; object_key; file_type; file_size; uploaded_by; created_at

Database rule: index all FKs plus common filter fields such as status, assigned_to/owner_id, due_date, required_date and created_at. Store timestamps consistently (recommended UTC) and convert to user/calendar timezone at the boundary.

4. Each Business Module — APIs + Tables

1. Director Daily Workspace

Purpose: Daily operating screen: calendar, due tasks, open actions, follow-ups, approvals, meeting preparation, communications, EOD summary.

Tables: daily_workspace_views, daily_workspace_items, daily_summaries

HTTP	Endpoint
GET	/api/v1/dashboard/daily
GET	/api/v1/dashboard/daily/items
GET	/api/v1/dashboard/daily/urgent
POST	/api/v1/dashboard/daily/summary

2. Calendar Management

Purpose: Calendar planning, scheduling, availability, conflicts, protected focus time, confirmations, links and reminders.

Tables: calendars, calendar_events, event_participants, availability_blocks

HTTP	Endpoint
GET	/api/v1/calendars
GET	/api/v1/calendar/events
POST	/api/v1/calendar/events
PUT	/api/v1/calendar/events/{id}
DELETE	/api/v1/calendar/events/{id}
GET	/api/v1/calendar/availability
POST	/api/v1/calendar/conflicts/check

3. Meeting Management

Purpose: Agenda/preparation, briefing, documents, attendance, minutes, decisions, actions and follow-up dispatch.

Tables: meetings, meeting_agendas, meeting_attendees, meeting_minutes, meeting_decisions, meeting_actions

HTTP	Endpoint
GET	/api/v1/meetings
POST	/api/v1/meetings
GET	/api/v1/meetings/{id}
PUT	/api/v1/meetings/{id}
POST	/api/v1/meetings/{id}/agenda
POST	/api/v1/meetings/{id}/minutes

POST	/api/v1/meetings/{id}/decisions
POST	/api/v1/meetings/{id}/actions
POST	/api/v1/meetings/{id}/dispatch-followups

4. Action Item Tracker

Purpose: Controls WHAT, WHO and WHEN from assignment through acknowledgement, progress, follow-up, escalation, closure and evidence.

Tables: action_items, action_updates, action_assignees, action_closure_evidence

HTTP	Endpoint
GET	/api/v1/actions
POST	/api/v1/actions
GET	/api/v1/actions/{id}
PUT	/api/v1/actions/{id}
POST	/api/v1/actions/{id}/acknowledge
POST	/api/v1/actions/{id}/updates
POST	/api/v1/actions/{id}/close

5. Decision Management

Purpose: Decision recording, context, informed people, related meetings/actions, implementation status and history.

Tables: decisions, decision_implementations, decision_stakeholders

HTTP	Endpoint
GET	/api/v1/decisions
POST	/api/v1/decisions
GET	/api/v1/decisions/{id}
PUT	/api/v1/decisions/{id}
POST	/api/v1/decisions/{id}/implementation
GET	/api/v1/decisions/{id}/history

6. Communication Management

Purpose: Email, WhatsApp/calls, internal/client communication, letters, replies, follow-ups, priority/tags and history.

Tables: communications, communication_participants, communication_threads, communication_messages

HTTP	Endpoint
GET	/api/v1/communications

POST	/api/v1/communications
GET	/api/v1/communications/{id}
PUT	/api/v1/communications/{id}
POST	/api/v1/communications/{id}/reply
POST	/api/v1/communications/{id}/followup
GET	/api/v1/communications/threads/{id}

7. Travel & Hospitality

Purpose: Travel requests, approvals, flight/train, hotel, transport, hospitality, itinerary/documents and expenses.

Tables: travel_requests, travel_bookings, travel_itineraries, travel_expenses, hospitality_requests

HTTP	Endpoint
GET	/api/v1/travel/requests
POST	/api/v1/travel/requests
GET	/api/v1/travel/requests/{id}
PUT	/api/v1/travel/requests/{id}
POST	/api/v1/travel/{id}/bookings
POST	/api/v1/travel/{id}/itinerary
POST	/api/v1/travel/{id}/expenses

8. Approval Management

Purpose: Approval requests, supporting documents, review, approved/changes/rejected, revisions and history.

Tables: approval_requests, approval_steps, approval_comments, approval_revisions

HTTP	Endpoint
GET	/api/v1/approvals
POST	/api/v1/approvals
GET	/api/v1/approvals/{id}
POST	/api/v1/approvals/{id}/submit
POST	/api/v1/approvals/{id}/approve
POST	/api/v1/approvals/{id}/request-changes
POST	/api/v1/approvals/{id}/reject
GET	/api/v1/approvals/{id}/history

9. Follow-up & Escalation

Purpose: Follow-up cycles, reminders, Level 1–5 escalation, overdue/director alerts and resolution tracking.

Tables: followup_cycles, escalation_logs, reminder_rules

HTTP	Endpoint
GET	/api/v1/followups
POST	/api/v1/followups
PUT	/api/v1/followups/{id}
POST	/api/v1/followups/{id}/complete
POST	/api/v1/escalations/run
POST	/api/v1/escalations/{id}/resolve
GET	/api/v1/escalations/history

10. Document & Records

Purpose: Controlled document storage, categories, confidential records, version control, access control and audit.

Tables: documents, document_categories, document_versions, document_access, document_tags

HTTP	Endpoint
GET	/api/v1/documents
POST	/api/v1/documents
GET	/api/v1/documents/{id}
POST	/api/v1/documents/{id}/versions
POST	/api/v1/documents/{id}/access
DELETE	/api/v1/documents/{id}
GET	/api/v1/documents/{id}/audit

11. Reports & Briefing Packs

Purpose: Daily/weekly summaries, meeting reports, KPI/MIS, decision summaries and leadership packs.

Tables: reports, report_schedules, report_filters, briefing_packs

HTTP	Endpoint
GET	/api/v1/reports
POST	/api/v1/reports
GET	/api/v1/reports/{id}
POST	/api/v1/reports/{id}/generate

GET	/api/v1/reports/kpi
POST	/api/v1/briefing-packs
GET	/api/v1/briefing-packs/{id}

5. Proposed Module Table Definitions

Table	Suggested columns / FK relationships
daily_workspace_views	id, user_id FK, view_date, timezone, layout_json, created_at, updated_at
daily_workspace_items	id, view_id FK, source_type, source_id, priority, sort_order, due_at, state
daily_summaries	id, user_id FK, summary_date, summary_text, urgent_count, overdue_count, generated_by, created_at
calendars	id, owner_user_id FK, calendar_name, timezone, provider, external_calendar_id, is_active
calendar_events	id, calendar_id FK, title, description, start_datetime, end_datetime, location, event_type, status, organizer_id FK
event_participants	id, event_id FK, user_id/contact_id, response_status, is_required
availability_blocks	id, user_id FK, block_type, start_datetime, end_datetime, reason, is_protected
meetings	id, calendar_event_id FK, title, purpose, meeting_date, location, organizer_id FK, status
meeting_agendas	id, meeting_id FK, agenda_text, version_no, prepared_by, status
meeting_attendees	id, meeting_id FK, user_id/contact_id, attendance_status
meeting_minutes	id, meeting_id FK, minutes_text, version_no, prepared_by, approved_at
meeting_decisions	id, meeting_id FK, decision_id FK
meeting_actions	id, meeting_id FK, action_item_id FK
action_items	id, source_type, source_id, title, description, assigned_to FK, due_date, priority_id FK, status, closed_at
action_updates	id, action_item_id FK, updated_by FK, progress_pct, update_text, next_due_date, created_at
action_assignees	id, action_item_id FK, user_id FK, is_primary, acknowledged_at
action_closure_evidence	id, action_item_id FK, attachment_id FK, submitted_by, submitted_at, remarks
decisions	id, meeting_id FK nullable, decision_text, context, made_by FK, decision_date, status
decision_implementations	id, decision_id FK, owner_id FK, target_date, status, completion_notes, completed_at
decision_stakeholders	id, decision_id FK, user_id/contact_id, informed_at, acknowledgement
communications	id, communication_type, subject, body, sender_id/contact_id, priority_id FK, status, sent_at
communication_participants	id, communication_id FK, participant_type, user_id/contact_id, direction
communication_threads	id, thread_key, subject, channel, status, last_message_at
communication_messages	id, thread_id FK, communication_id FK, message_body, sender_id, received_at
travel_requests	id, requester_id FK, purpose, from_location, to_location, start_date, end_date, traveller_count, status
travel_bookings	id, travel_request_id FK, booking_type, provider, booking_reference, departure_at, arrival_at, cost, status
travel_itineraries	id, travel_request_id FK, itinerary_version, itinerary_text, prepared_by, status

travel_expenses	id, travel_request_id FK, expense_date, category, amount, currency, receipt_attachment_id FK, status
hospitality_requests	id, travel_request_id FK nullable, event_date, venue, guest_count, requirement_text, budget, status
approval_requests	id, module_type, reference_id, requester_id FK, title, reason, priority_id FK, status, submitted_at, decided_at
approval_steps	id, approval_request_id FK, step_no, approver_id FK, status, decided_at, remarks
approval_comments	id, approval_request_id FK, user_id FK, comment_text, created_at
approval_revisions	id, approval_request_id FK, revision_no, changed_by FK, change_summary, submitted_at
followup_cycles	id, reference_type, reference_id, owner_id FK, cycle_no, sent_at, expected_response_at, completed_at, status
escalation_logs	id, reference_type, reference_id, escalation_level_id FK, escalated_to FK, reason, escalated_at, resolved_at, status
reminder_rules	id, reference_type, reference_id, trigger_type, offset_minutes, recipient_rule, is_active
documents	id, category_id FK, title, description, file_type, object_key, file_size, checksum, uploaded_by FK, is_confidential, created_at
document_categories	id, category_name, retention_days, confidentiality_level, is_active
document_versions	id, document_id FK, version_no, object_key, checksum, file_size, uploaded_by FK, uploaded_at
document_access	id, document_id FK, user_id FK, access_type, expires_at, granted_by FK
document_tags	id, document_id FK, tag_name
reports	id, report_name, report_type, query_definition, created_by FK, status, generated_at
report_schedules	id, report_id FK, frequency, next_run_at, recipients_json, is_active
report_filters	id, report_id FK, filter_key, filter_value
briefing_packs	id, title, pack_type, period_start, period_end, created_by FK, status, generated_at

6. Common Workflow & Escalation Design

Stage	Meaning / implementation
Captured	Record created through intake or directly by module.
Information Checked	Required details, attachments, priority and due date validated.
Planned	Placed into calendar/task/operational plan.
In Progress	Owner executes and records progress.
Waiting / Follow-up	Dependency + response owner + expected response + next follow-up stored.
Submitted	Output/update submitted.
Review	Director/EA review where required.
Revision / Rework	Revision number/date and reviewer remarks retained; same record resubmitted.
Completed	Accepted, closure evidence captured where required.
Archived	Closed history remains queryable.
Analytics	Operational facts feed dashboards/KPIs/insights.

Level	Rule
1	EA reminder when action is overdue
2	Escalate to relevant Department Head
3	Director alert for high-priority/materially overdue work
4	Urgent Director decision/intervention
5	Critical confidential/legal/client/emergency matter — immediate escalation

7. API Standards & Background Jobs

Topic	Recommendation
API versioning	Use /api/v1/... from first production release.
Auth	JWT + RBAC; resource-level authorization for confidential records.
State changes	Transactional service methods; reject invalid transitions with 409/422.
Validation	DTO validation at API boundary + business/domain validation in service layer.
Files	Upload binaries to Spaces; MySQL stores metadata/object_key/checksum/access.
Audit	Append audit record for create/update/delete/status/revision/approval/escalation/access events.
Pagination	Page/cursor all operational list APIs; standardize filters and sorting.
Idempotency	Use Idempotency-Key for duplicate-prone POST operations.
Errors	400 validation; 401 auth; 403 permission; 404 missing; 409 state conflict; 422 business rule; 500 unexpected.
Jobs	Due-date monitor; follow-up dispatcher; escalation evaluator; notification dispatcher; recurring work generator; report generator.

8. AI Integration Boundary

AI is an assistance layer, not a separate business module. Recommended pattern: **API** → **Application Service** → **AI Orchestrator** → **model/provider** → **structured suggestion** → **human confirmation** → **business transaction** → **audit**. This matches the source's positioning of AI across intake, daily planning, meetings, communication, decisions, follow-up, revision and reporting. █filecite█turn0file0█L238-L247█

9. Security & Go-Live Checklist

- RBAC with module + record-level authorization for confidential records.
- Private/expiring file URLs and explicit document access grants.
- Immutable audit/history records and searchable operational history.
- DB indexes on FKs and common filters; optimistic concurrency for concurrent edits.
- HTTPS/TLS, secret management, validation, file-type/size checks and rate limiting.
- MySQL backup + tested restore; Spaces protection/retention strategy.
- Unit + integration + e2e tests; workflow transition tests; escalation timing tests.
- Load test dashboard/report queries and background jobs before production.
- Swagger/OpenAPI reviewed with frontend consumers; consistent DTOs and error contracts.
- UAT sign-off against each functional item in the source specification.

10. Definition of Done — Every Module

For every module, complete: **migration** → **entity/model** → **repository/data access** → **service/domain rules** → **DTOs/validation** → **controller/API** → **authorization** → **workflow integration** → **audit/history** → **attachments where required** → **notification hooks** → **unit/integration tests** → **Swagger/OpenAPI** → **Angular API client/UI integration** → **UAT**.